

The effect of low-flow sevoflurane and desflurane on pulmonary mechanics during laparoscopic surgery.

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This study was designed to investigate the effects of inhaled anesthetic agents on respiratory mechanics with low flow anesthesia in laparoscopic abdominal surgery. Two inhaled anesthetics, desflurane and sevoflurane, have a lower solubility in blood and tissues than all previous volatile anesthetics, and have become the preferred volatile anesthetics for routine surgical procedures. Twenty-six patients were examined. Patients were randomly assigned to two groups, to receive sevoflurane ($n = 13$) or desflurane ($n = 13$). Tidal volume and ventilation rate were kept unchanged throughout the operation. Intra-abdominal pressure was kept constant at the level of 12 mm Hg. Respiratory mechanics such as peak inspiratory pressure (PIP), respiratory resistance (Rr), and dynamic compliance (Cdyn) measurements were recorded by a Datex-Ohmeda respiratory device (Datex-Ohmeda, Finland) at four timepoints: 5 minutes after mechanical ventilation started (T1), after insufflation of the peritoneum (T2), in the 30-degree Trendelenburg position (T3), and after desufflation of the peritoneum (T4). In our study, desflurane caused a statistically significant increase in PIP and Rr and decrease in C(dyn). When the two groups were compared, Rr values in the desflurane group showed significant increases at T2, T3, and T4 compared to the sevoflurane group ($P < 0.05$). These values did not change in the sevoflurane group, while PIP significantly increased at T2, T3, and T4 after desufflation in the desflurane group ($P < 0.05$). Cdyn values decreased significantly at all 4 timepoints in the desflurane group compared to the sevoflurane group ($P < 0.05$). We concluded that respiratory mechanics were affected by desflurane with low flow anesthesia in patients undergoing laparoscopic abdominal surgery. No significant influence on respiratory mechanics was seen with sevoflurane anesthesia.

Volatile Anesthetics Regulate Anti-Cancer Relevant Signaling.

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Volatile anesthetics are widely used inhalation anesthetics in clinical anesthesia. In recent years, the regulation of anti-cancer relevant signaling of volatile anesthetics has drawn the attention of investigators. However, their underlying mechanism remains unclear. This review summarizes the research progress on the regulation of anti-cancer relevant signaling of volatile anesthetics, including sevoflurane, desflurane, xenon, isoflurane, and halothane in vitro, in vivo, and clinical studies. The present review article aims to provide a general overview of regulation of anti-cancer relevant signaling and explore potential underlying molecular mechanisms of volatile anesthetics. It may promote promising insights of guiding clinical anesthesia procedure and instructing enhance recovery after surgery (ERAS) with latent benefits.

Pulmonary vasodilator response to adenosine triphosphate-sensitive potassium channel activation is attenuated during desflurane but preserved during sevoflurane anesthesia compared with the conscious state.

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The objective of this study was to investigate the effects of sevoflurane and desflurane anesthesia on the

pulmonary vasodilator response to the adenosine triphosphate-sensitive potassium channel agonist, lemakalim, compared with the response measured in the conscious state. In addition, the authors assessed the extent to which sympathetic α_1 -adrenoreceptor inhibition and cyclooxygenase pathway inhibition modulate the vasodilator response to lemakalim. Twenty-four conditioned male mongrel dogs were chronically instrumented to measure the left pulmonary vascular pressure-flow relationship. After precontraction with the thromboxane analogue, U46619, dose-response relationships to lemakalim were assessed on separate days in the conscious state and during sevoflurane (approximately 3.5% end-tidal) and desflurane (approximately 10.5% end-tidal) anesthesia (approximately 1.5 minimum alveolar concentration for each anesthetic agent). The effects of sympathetic α_1 -adrenoreceptor inhibition (prazosin) and cyclooxygenase inhibition (indomethacin) on the pulmonary vasodilator response to lemakalim also were assessed in the conscious and desflurane-anesthetized states. Neither sevoflurane nor desflurane had a net effect on the baseline left pulmonary vascular pressure-flow relationship compared with the conscious state. The magnitude of the pulmonary vasodilator response to lemakalim was preserved during sevoflurane anesthesia but was attenuated ($P < 0.05$) during desflurane anesthesia compared with the conscious state. The attenuated lemakalim-induced vasodilator response during desflurane anesthesia was partially reversed ($P < 0.05$) by pretreatment with prazosin but not indomethacin. These results indicate that adenosine triphosphate-sensitive potassium channel-mediated pulmonary vasodilation is preserved during sevoflurane anesthesia but is attenuated during desflurane anesthesia. The attenuated response to adenosine triphosphate-sensitive potassium channel activation during desflurane anesthesia is partially mediated by reflex sympathetic α_1 -adrenoreceptor vasoconstriction.

Monitoring of liver and kidney profiles in anesthesiologists working in a regional reference teaching hospital in Northern Italy: analysis of health surveillance data using a linear mixed model.

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Anesthesiologists represent an occupational group exposed to specific occupational hazards, including potential exposure to waste anesthetic gas released during medical procedures. In recent decades, halogenated anesthetic gases, such as desflurane and sevoflurane, have largely replaced nitrous oxide, due to better safety profiles and lower adverse health effects. However, possible long-term effects of low concentration exposures are unknown. A longitudinal analysis of health surveillance data was performed to test for possible changes over time in key markers of liver and kidney function. Moreover, we assessed the appropriateness of applying linear mixed models to occupational health data. A retrospective cohort study was conducted using health surveillance data from a cohort of anesthesiologists and a cohort of unexposed physicians working at the Polyclinic Hospital San Martino of Genoa, Italy, during 2016-2022. A 2-level linear mixed model with covariance structure of first order autoregressive model (AR(1)) type at the first level and unstructured type at the second level was applied. One hundred seventy subjects were included in the analysis, equally divided between exposed and unexposed. At the first and last periodic examination, liver and kidney markers were not statistically different in the 2 cohorts. The only significant change found related to estimated glomerular filtrate, which was found at the last follow-up to be greater among the exposed ($M = 104.18$ vs. 90.07 , $p = 0.007$). The linear mixed model showed that anesthetic gas exposure was not associated with any of the outcomes. These results suggest the absence of increase in liver and kidney profile markers in the study population. Health surveillance data, aggregated and analyzed with appropriate statistical models, allow inferences to be made about potential health effects of workers due to uncontrolled exposures. To this end, the linear mixed model

represents a powerful tool for longitudinal analysis of data derived from monitoring workers. *Int J Occup Med Environ Health*. 2024;37(5):557-68.

Effects of volatile anaesthetics on incidence of postoperative depression and anxiety symptoms in elderly patients: A retrospective analysis of a prospective cohort study.

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Studies have suggested volatile anaesthetics may alleviate depression and anxiety symptoms in patients. However, there is a paucity of research in this area. We wanted to determine the association between volatile anaesthetics and depression/anxiety symptoms in elderly patients within 7 days after surgery. This study retrospectively analysed data from a prospective database of patients aged 65 and above who underwent non-cardiac, non-neurosurgical elective surgery in 19 tertiary hospitals across 10 provinces in China. The study period spanned from April 1, 2020 to April 30, 2022. Patients receiving volatile anaesthetics received at least one volatile anaesthetic (sevoflurane, isoflurane, desflurane), and those who received non-volatile anaesthetics did not receive any volatile anaesthetic. Binary logistic regression analyses were conducted, and propensity score-matching (PSM) and subgroup analyses were also applied. A total of 7165 patients were included in the analysis. Among them, 4957 (69.18%) cases received volatile anaesthetics. The administration of volatile anaesthetics was associated with a lower rate of postoperative depression [22.76% vs. 27.26%, odds ratio (OR): 0.75, $P < 0.001$], and reduced the incidence of postoperative anxiety (19.59% vs. 24.68%, OR: 0.78, $P < 0.001$). The risk of postoperative depression (24.78% vs. 27.93%, OR: 0.85, $P = 0.035$), and anxiety (21.45% vs. 25.92%, OR: 0.81, $P = 0.006$) were both significantly decreased in the volatile anaesthetics group in the PSM cohort. The results suggest that using volatile anaesthetics maybe associated with a reduction in the incidence of depression and anxiety in elderly patients during the early postoperative period.

Effect of desflurane, sevoflurane or propofol on the incidence of postoperative delirium in older adults undergoing moderate- to high-risk major non-cardiac surgery: study protocol for a prospective, randomised, observer-blinded, clinical trial (RAPID-II trial).

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The effect of different anaesthetics on the incidence of postoperative delirium is still not entirely clear. Therefore, we will evaluate the effect of desflurane versus sevoflurane versus propofol for the maintenance of anaesthesia on the incidence of postoperative delirium in older adults undergoing moderate- to high-risk major non-cardiac surgery. We will further compare the incidences of delayed neurocognitive recovery, long-term postoperative neurocognitive disorder, postoperative nausea and vomiting between the groups. In this multicentre, prospective, observer-blinded, randomised controlled clinical trial, we will include 1332 patients ≥ 65 years of age undergoing moderate- to high-risk major non-cardiac surgery lasting at least 2 hours. Patients will be randomly 1:1:1 assigned to receive desflurane, sevoflurane or propofol for anaesthesia. Maintenance of anaesthesia will be performed in a goal-directed manner using processed electroencephalography with an intraoperative goal of bispectral index 40-60. Our primary outcome will be the incidence of postoperative delirium within the first five postoperative days. Postoperative delirium will be assessed using the three-dimensional-confusion assessment method (3D-CAM) or CAM-intensive care unit (ICU) in the morning and evening of the first

five postoperative days by blinded study personnel. The primary outcome, the incidence of postoperative delirium, will be compared between the three study groups using a χ^2 test. Furthermore, a logistic regression model for the incidence of postoperative delirium will be performed, accounting for randomised groups as well as other predefined confounding factors. This clinical trial has been approved by the ethics committee and the Federal Office for Safety in Healthcare as the competent authority for clinical trials in Austria. The results of this trial will be published in a peer-reviewed journal. ClinicalTrials.gov NCT05990790.

Anesthetic Management of a Pediatric Patient With Pfeiffer Syndrome.

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Pfeiffer syndrome (PS) is a rare inherited disorder that affects the craniofacial and extremity bones but can also include anomalies of the upper and lower airways. Among the 3 subtypes, PS types 2 and 3 are the most severe. We describe the anesthetic management of a 10-year-old boy with PS type 2 scheduled for dental treatment under general anesthesia. Because of his history of Chiari malformation, avoiding neck hyperextension was recommended to prevent worsening of his neurologic function. Preoperative computed tomography revealed significant nasal stenosis but no tracheal anomalies. Considering the difficulty of nasotracheal intubation, we planned for an oral intubation. General anesthesia was induced using sevoflurane, nitrous oxide, and oxygen along with an oropharyngeal airway. After lidocaine was topically applied to the oropharynx, fiber-optic orotracheal intubation was performed under spontaneous ventilation to minimize head and neck movement. Anesthesia was maintained using desflurane and remifentanyl, and no postoperative complications were observed. General anesthesia for patients with PS requires careful planning, which should include preoperative assessment of the airway to determine the feasibility of nasotracheal intubation and identify airway irregularities.

Anesthesia and its environmental impact: approaches to minimize exposure to anesthetic gases and reduce waste.

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In today's era of modern healthcare, the intersection between medical practices and environmental responsibility has gained significant attention. One such area of focus is the practice of anesthesia, which plays a crucial role in various surgical procedures. Anesthetics such as nitrous oxide and volatile halogenated ethers (desflurane, isoflurane, sevoflurane) are examples of medical gases that are strong greenhouse gases that contribute to global warming. During medical procedures, most of these anesthetic agents are released into the atmosphere, which exacerbates their influence on the environment. Also anesthesia delivery systems have traditionally utilized high flow rates of gases, leading to not only excessive consumption but also a considerable environmental impact in terms of greenhouse gas emissions. However, the emergence of low-flow anesthesia (LFA) presents a promising solution for achieving emission reduction and cost savings, thereby aligning healthcare practices with sustainability goals. Understanding LFA involves the administration of anesthetic gases to patients at reduced flow rates compared to conventional high-flow methods. This practice requires precision in gas delivery, often incorporating advanced monitoring and control systems. By optimizing gas flow to match the patient's requirements, LFA minimizes wastage and excessive gas release into the environment, subsequently

curbing the carbon footprint associated with healthcare operations. Decreasing volatile anesthetic delivery provides safe and effective strategies for anesthesia providers to decrease costs and reduce environmental pollution. Current literature support in favor of LFA represents an area of cost containment and an opportunity to lessen the environmental impact of anesthesia. This article will cover the concept of LFA, the distinctions between low flow and minimal flow, and the potential advantages of LFA, such as those related to patient safety, the environment, and the economy.

Comparative effect of desflurane and sevoflurane on liver function tests of patients with impaired hepatic function undergoing cholecystectomy: A randomized clinical study.

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Desflurane and sevoflurane are the most common volatile anesthetics used during laparoscopic and hepatic surgery. The objective of the study was to evaluate the effect of desflurane and sevoflurane in patients with elevated preoperative liver functions undergoing laparoscopic cholecystectomy. The study was a randomized study and included 162 patients classified randomly into two groups: Desflurane group: The patients received desflurane (end-tidal concentration 4%-6%) as an inhalational agent during the whole procedure. Sevoflurane group: The patients received sevoflurane (end-tidal concentration 2%-4%) as an inhalational agent during the whole procedure. The investigations included serum level of aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase (ALP), gamma-glutamyltransferase (GGT), and total bilirubin. The values were serially collected at the following timepoints; T0:at the preoperative period, T1:directly after surgery, T2:1st postoperative day, T3:2nd postoperative day, T4:3rd postoperative day, T5:5th postoperative day, T6:7th postoperative day, and T7:10th postoperative day. The statistics were described in terms of mean $\pm$ standard deviation, frequencies, and percentages. The preoperative liver enzymes and total bilirubin were higher than the normal range in patients of the two groups. Postoperatively, there was a decrease in the AST and ALT with desflurane more than sevoflurane from T1 to T6($P < 0.05$). The ALP, GGT, and bilirubin decreased in patients of the two groups, but the comparison was insignificant ($P > 0.05$). The desflurane is a safe inhalational volatile for maintenance of anesthesia in patients with impaired liver function undergoing laparoscopic cholecystectomy. It was associated with a decrease in the liver enzymes more than the sevoflurane.

The mechanisms of cardio-protective effects of desflurane and sevoflurane at the time of reperfusion: anaesthetic post-conditioning potentially translatable to humans?

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Myocardial conditioning is actually an essential strategy in the management of ischaemia-reperfusion injury. The concept of anaesthetic post-conditioning is intriguing, its action occurring at a pivotal moment (that of reperfusion when ischaemia reperfusion lesions are initiated) where the activation of these cardio-protective mechanisms could overpower the mechanisms leading to ischaemia reperfusion injuries. Desflurane and sevoflurane are volatile anaesthetics frequently used during cardiac surgery. This review focuses on the efficacy of desflurane and sevoflurane administered during early reperfusion as a potential cardio-protective strategy. In the context of experimental studies in animal models and in human atrial tissues in vitro, the mechanisms underlying the cardio-protective effect of these agents and

their capacity to induce post-conditioning have been reviewed in detail, underlining the role of reactive oxygen species generation, the activation of the cellular signalling pathways, and the actions on mitochondria along with the translatable actions in humans; this might well be sufficient to set the basis for launching randomized clinical studies, actually needed to confirm this strategy as one of real impact.
